

Comparison of PPO and SAC Algorithms Towards Decision Making Strategies for Collision Avoidance Among Multiple Autonomous Vehicles

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Abstract—Multiple vehicle collision avoidance strategies with safe lane changing strategy for vehicle control using learning base technique are the most crucial concern in autonomous driving system. Statistics shows that the latest autonomous driving systems are usually prone to rear-end collision. Rear-end collisions often result in severe injuries as well as traffic jam and the consequences are much worse for multiple-vehicle collision. Many previous autonomous driving research focused solely on collision avoidance strategies for two consecutive vehicles. This study proposes a centralised control strategy for multiple vehicles using reinforcement learning focused on partner consideration and goal attainment. The system depicted as a group of vehicles are communicate and coordinate each others by a set of rays and maintain a short following move away. In order to address this challenge, a simulation was implemented in the Unity3D game engine and two state-of-the-art RL algorithms PPO (Proximal Policy Optimization) and SAC (Soft Actor-Critic) were trained by an agent using Unity ML-Agents Toolkit. In terms of success rate, performance, training speed and stability two algorithms are comparable. The potency of algorithms has been assessed by the traffic flow (1) change in vehicle speed, (2) differ in the vehicle beginning positions, and (3) switch to next lane. The agent performed similarly at a 91% success rate in PPO or SAC applications

Keywords— Autonomous driving, Multiple vehicle collision, Robotics, Reinforcement Learning.

I. INTRODUCTION

Road traffic incidents are widely regarded as one of the world's biggest social issues. On highway conditions, rear-end crashes accounted for 42.7 percent of all accidents [1], [2]. The first crash caused by rear-end typically induces multiple crashes including various examples of severe accidents involving more than 100 vehicles. The market penetration of self-driving cars has increased in recent years. An autonomous vehicle's driving plan strategy completes three jobs, including mission planning, where the vehicle pops a routing problem to fulfil a task, decision-making where the vehicle selects a suitable action for the ensuing next step from available action planning, and path planning, in which the vehicle plans its upcoming trajectory in terms of time and space. Multi agent self-governing exploration is an influential submission of real world multi-vehicle

systems, where a bunch of connected robots are coordinated to delve into any unknown situation collaboratively. This method has recently gained a major research attention because of its utility in exploration and rescue, identification and tracking of faults, location and mapping, etc. This paper deals with the subject of autonomous driving planning. We specifically endeavor to reach an expert driver's decision making, i.e., by taking into account the stochastic driving behavior of natural expert driving policies, we want to imitate the optimum driving approach involving various distinctive driving actions such as lane changing, lane keeping, speed maintenance, braking and acceleration. In [3] Sugiyama et al. draw from the optimal velocity model's collision criterion and investigated the factors that influence multiple-vehicle collisions. *Figure:1* shows the consequences of multiple vehicle collision by abrupt lane change. *Z. Li et al.* looked at the prospect of a multiple-vehicle collision caused by an abrupt halt while taking into account the impact of the intelligent control mechanism [4]. *Nagatani et al.* [5] investigated the impact of the friction force of velocity-dependent on the multiple-vehicle collision. Regarding these research work we proposed a learning based threat assessment frame work to predict the multiple vehicle collision in [6].

Investigations shown that in the majority of severe rear-end collision the following vehicles driver uses insufficient or ineffective braking [7]. In contrary, abrupt lane changes or braking by the leading vehicle will leave the following vehicle with inadequate time to react, resulting in collision. Current vehicle control systems are centered on traditional theory of optimum control, which defines the control input usually over a particular period of time and limits it to a viable space [8]. The control methods of velocity consist of closed loop control processes such as PID (Proportional Integral Derivative) and MPC (Model Predictive Control) controllers [9]. PID controllers are employed to minimize the cost function cost that has three terms: the proportional term as current error, the integral term as effect of past errors, and the derivative term as effect of future errors.

Moreover, conventional control strategies, i. e., the CSP (Constant Spacing Policy), the CTHP (Constant Time Headway Policy), and the SMC (Sliding Mode Control) [10], have underprivileged adaptability to environment and are incapable of making accurate and optimal decisions in complex and diverse driving environments, particularly when AVs and human driving vehicles co-exist. The intelligent control system, which is aided by current Artificial Intelligence (AI)

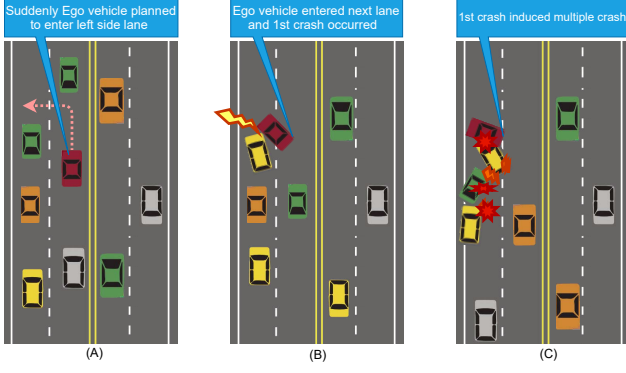


Figure 1: *Evaluation of multiple collision caused by sudden lane change where in 1(A) ego vehicle attempt to change ego lane to left side lane, 1(B) the ego vehicle changed lane and the 1st crash occurred, and 1(C) 1st collision induced the multiple collisions.*

technology development [11], not only creates judgments centered on the existing environment, but also constantly learns and acclimates to it. Reinforcement Learning (RL)-centered autonomous decision-making systems function pretty well due to this ongoing learning and feedback process. In the deep reinforcement learning method, the agent's observations are sent into a neural network that produces an action depending on its present policy. These methods are highly susceptible to tuning hyper parameters like the learning rate and the initialization. PPO addresses these concerns by being easily tuned and easy to implement. As an on-policy gradient approach it learns directly from the data which collects from the current states of its environment, not the process of Q-learning like deep Q-Networks, learning process is depends on previously collected offline data. In contrast, SAC which has an off-policy technique, means that the learning process depends on a buffer stored data from past episodes. SAC thereby utilizes the experience of earlier episodes, whereas the agent is only present experience in on-going policies.

Multiple vehicle and multi-lane traffic scenarios pose complex issues due to limited navigation and unpredictable vehicle interactions [12].

The following are the key concerns of this paper:

1. For the formulation of the collision avoidance and impact mitigation problem, coupled multiple vehicles (four vehicles) are directly taken into account.
2. By introducing a lower bound as a constraint of the leader vehicle's deceleration, vehicles in front of the leader vehicle of the coupled group have been taken into account.
3. By introducing an upper limit to the deceleration of the final vehicle in the coupled group, vehicles further behind the coupled group have been taken into account as well. This is a usual scenario with mixed traffic and increasing market penetration of V2V [13] and autonomous vehicle control [14]. *Figure:2* shows the illustration of successful mechanism of avoiding multiple vehicle collision by lane change or lane merging problem.

The rest of the paper is structured accordingly. In Section II, the fundamentals of PPO and SAC algorithms and the core comparison concerns are depicted. The associated work is provided in section III. In methodology details, the simulation setup, training methods are presented along with the comprehensive discussion on simulation results in section IV, and Section V draws the conclusion.

II. REINFORCEMENT LEARNING ALGORITHMS

Due to comparing the performance of these two algorithms we should concern the basic mechanism of PPO which is defined by two parts, the policy gradient loss express by *equation 1* and the trust region express by *equation 2*.

$$\mathcal{L}^{pg}(\theta) = \hat{\mathbb{E}}_t \left[\log \pi_{\theta}(\alpha_t | s_t) \hat{B}_t \right] \quad (1)$$

Where, $\pi_{\theta}(\alpha_t | s_t)$ has indicated the α_t taken action in t time given s_t state with π_{θ} policy parameters. The loss function $\mathcal{L}^{pg}(\theta)$ is the estimated reward of the action at time t , $\hat{\mathbb{E}}_t$, and a second term, $\hat{B}_t$ the function which represents the relative cost of the carefully chosen action at time t . TRPO ensures that the Policy Optimization does not go too far from the previous policy while revising the policy.

$$\text{maximize}_{\theta} \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(\alpha_t | s_t)}{\pi_{\theta_{pre}}(\alpha_t | s_t)} \right] \hat{B}_t$$

$$\text{subject to } \hat{\mathbb{E}}_t [FL[\pi_{\theta_{pre}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)]] \leq \delta \quad (2)$$

The action chosen for the policy is called $\pi_{\theta}(\cdot | s_t)$ as the policies are stochastic and not deterministic. In order to avoid the over-fitting of the policy, a constraint on the FL is added to the objective of optimization and the amount of change to the policy is limited. The FL constraint, on the other hand, restricts the optimization procedure and may occasionally lead to unwanted training behavior. PPO contains this additional constraint directly in order to minimize this problem [15].

III. RELATED WORKS

Multi-agent reinforcement learning has recently attracted interest by tackling several issues, including autonomous vehicles [16], [17]. The recent advances in artificial neural network and deep reinforcement learning that consents multiple agents in different environments to be studied, multi-agent reinforcement learning has prospered. In existing multi agent studies, there are two types of tasks: cooperative and competitive, both of which aim to co-evolve more than two agents who cooperate with each other parallelly in the environment. Some times in multi-agent systems, Single agent methods may be directly used with slight changes [18]. One easiest way is to get the optimal rewards from their individual training where the partners agents are consider as a part of environment [19], [20]. However, this strategy challenges the basic premise that reinforcement learning should be steady and Markovian in the environment.

Several modelling methods have been suggested to solve the autonomous sudden lane change and lane merging problem. One solution was a slot-based algorithm that identified either a free or occupied slot occupancy state centered on the agents' acceleration and decline speed, position and driving behavior [21]. For decision-making on the topic of lane merging of an urban expressway, a theory regarding driving regulations and accepting gaps was developed [22]. In opposite to rule based models, machine learning models may handle without resorting to predetermined rules or models [23] with complex road traffic conditions. In particular, reinforcement learning can learn best by itself via trial and error. A trained reinforcement agent can monitor its surroundings, interacts with its surroundings by means of specified activities and is rewarded or penalized for these interactions based on the target state. The agent will select the best policy to accomplish its goal by boosting the reward, and therefore will merge effectively. Using deep Q-learning, reinforcement learning systems have already achieved some success story in merge challenge [22]. Two modern enhancement algorithms are being used in the current investigation. There are two reasons why reinforcement learning methods can be used in order to solve the lane changing and merging problem and compare both two techniques. The first one is PPO (Proximal Policy Optimization), presented by the team OpenAI in 2017 and it has huge success story in very

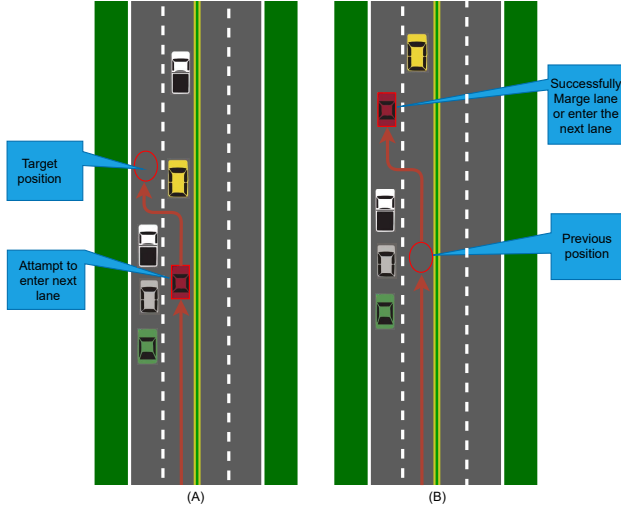


Figure 2: Illustration of avoidance mechanism of multiple vehicle collision

difficult environments like MOBA Dota2, which is one of the most sophisticated and dominant E-sport games to date [24]. We compared this algorithm with the SAC (Soft Actor-Critic) algorithm that is an off policy reinforcement learning method depends on actor-critic framework. We have a continuous action problem and the decision making process is also continuous for a time being, so we select these two algorithms PPO and SAC. PPO seems easier to implement in comparison to other on-policy reinforcement algorithms and at least is in accordance with or exceeds the same reinforcement algorithms of learning. Although SAC is not so simple to implement, it can overdo other policies and off-policy methods by combining off-policy updates with a stable stochastic actor-critical formula. Compared to other on-policy reinforcement learning algorithms, PPO seems to be easier to implement and is at least on par or outperforms similar reinforcement learning algorithms [24], [25]. While SAC is not as easily implemented, by combining off-policy updates with a stable stochastic actor-critic formulation, it could outperform other on-policy and off-policy methods [26].

IV. METHODOLOGY

A. Agent

The agent contains with two features first one is goal attainment and second one is partners consideration. In this way it will perform successful lane change and will be able to avoid the abrupt lane change situation. The elaborately criteria of successful collision avoidance are (a) no collision with other road users and also roadside infrastructures, (b) stopping inside lane is prohibited, although it is okay to slow down, and (c) reaching the road end shows at top of screen. The agent will be awarded at the ending of episode for a successful merge, also rewarded by crossing checkpoints along its journey and is awarded according to its speed. The agent shall be penalized if a merger fails. For every decision taken by the agent, a tiny punishment is also imposed to make sure that decisions are made most efficient, leading eventually to straight away driving. While most advanced autonomous vehicle use effective sensor as LiDAR, in this setup only two set of rays are being used. First set of rays indicates the distance from the sides of the lane. The other second set is to detect the distance to other vehicles around the ego vehicle. Furthermore, the agent understands its own speed and its steer direction, so that it may drive the vehicle according to the speed.

B. Simulation setup

A simulation has been developed in order to resolve the problem of lane change, to enable the reinforcement learner to successfully merge of trained agent with SAC or PPO. Simulation was created using Unity3d game engine with an integrated physics engine that enables for realistic car controls including wheel spinning and as well as colliding with the terrain, suspension of wheels, slip and more. Its ML-Agents Suite is used to support the neural network, which allows the development and formation of machine learning models through the TensorFlow library, an open-source library. This 3D simulation allows that the car to undergo a gravitation, dredging and rolling impact. Figure:3 is the training phase presentation of SAC algorithm.

C. Comparison

This section the driving efficiency, learning rate, and overall performance of the two methods are compared. Data is acquired by having the agents perform 300 episodes to compare the two methods. The overall success rate of the agents determines performance, the overall quantity of steering and the average timesteps each episode are used to calculate efficiency and Evaluating the robustness of these two algorithms all others road participant car has different starting point and speeds ranging from 80 to 120 km/h. Sensors are installed in all vehicles to detect the presence of another vehicle. If it gets empty, it has a random opportunity to merge on the unoccupied lane. When it's a car on the front, the car slows down. It will speed up when a car is behind the car. If a car is on both the front and the back, the vehicle makes sure the same distance between the cars is retained.

D. Results

The data shows in Table 1, its reveal that the two algorithms have done the same. With at least a 91% success rate, the agent can merge on the highway. When cars are close, if the agent attempts to merged it slows down or accelerates, depending on the proximity to the road and the proximity of the other participants, which is preferable. The two algorithms have proceeded until they have either done 3,000,000 times or training for 20 hours. The episode's maximum reward is 66.2, therefore on average 60 are regarded decent. While SAC required lower times than PPO, approximately 6.6 million steps compared to 1.4 million steps, the average decent reward compared to the 4 hours of PPO training required about 12 hours. The PPO's time increment was more stable than the SAC and its performance fluctuations were lower once a high success rate, shown by the quick decline in performance of 1.6 million timestep. The number of times per episode is identical to that of the two algorithms; however, the SAC agent had roughly double the steering amount per episode. Although both agents completed the episode in a relatively equal number of actions. Although for the two algorithms, the timesteps are identical per episode, suggesting that both agents performed the episode with the same number of actions, the SAC agent roughly double the number of steering fractions every episode. Illustration of both algorithms' performance shown in figures: 4 and 5.

E. Discussion

This work deploys the multiple vehicle collision avoidance strategies in autonomous driving and compares the two RL algorithms PPO and SAC by enabling an agent to learn how to perform this. SAC and PPO performed equivalent in terms of total performance, respectively with a success rate of 91% and 96%. The SAC algorithm required less time and fewer episodes in terms of learning rates than the PPO to achieve this rate of success, suggestive of their sample efficiency. However, the processing of these episodes required SAC more time than PPO, probably because of SAC's off-policy updates. In this context, SAC over PPO should be used when there is a restricted sample supply, as it learns from all prior experiences. PPO trains more

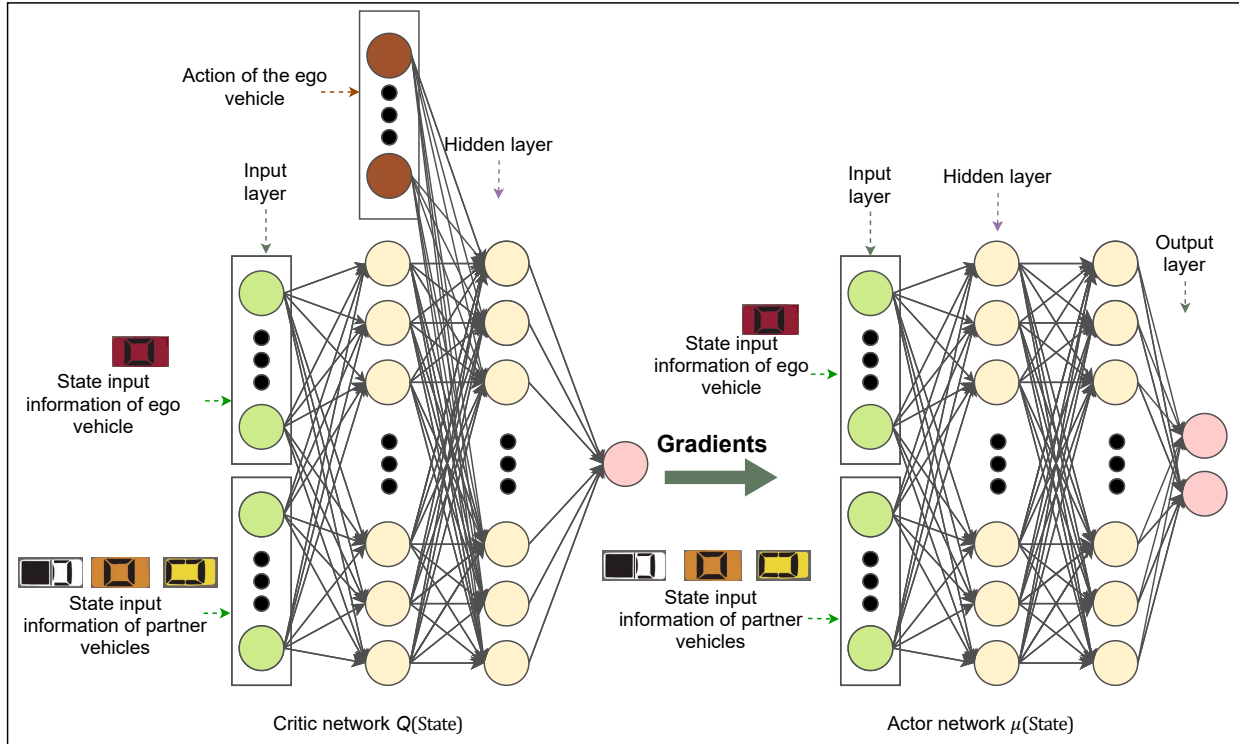


Figure 3: Training phase of SAC algorithm

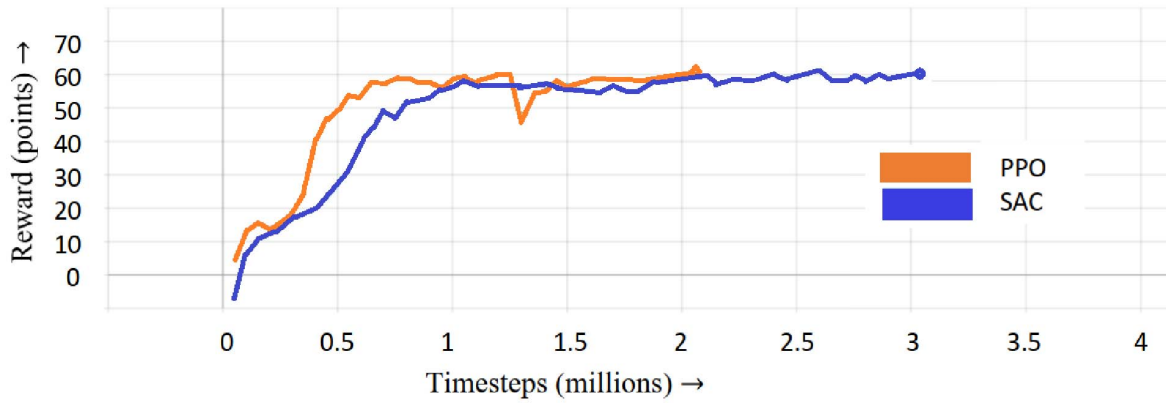


Figure 4: For each algorithm, the average episode reward against actions

rapidly but needs more data to succeed. In terms of driving efficiency, the SAC agent took more timesteps to complete an episode and had more total amount of steering. PPO took about 140 timesteps while SAC took 160 timesteps to successfully merge, indicating that SAC took more inefficient actions compared to PPO. There was also a large difference between the total amount of steering done by the agents. Thus, the SAC agent took more unnecessary actions and worse actions on average, making it sub optimal compared to PPO in this regard. However, the problem of over fitting does emerge with both SAC and PPO as this also occurs in most machine learning problems. Table: I present the performance information of two algorithms.

Table I: Evaluation of Algorithm Performance

Algorithm	PPO	SAC
Success rate	91%	94%
Timesteps till 60 rewards	2 million	660,000
Timesteps	3 million	2 million
Time till 60 rewards	5 hours	9 hours
Training hours	12	20
Per episode avg steering change	46	93
Per episode avg timesteps	55	175

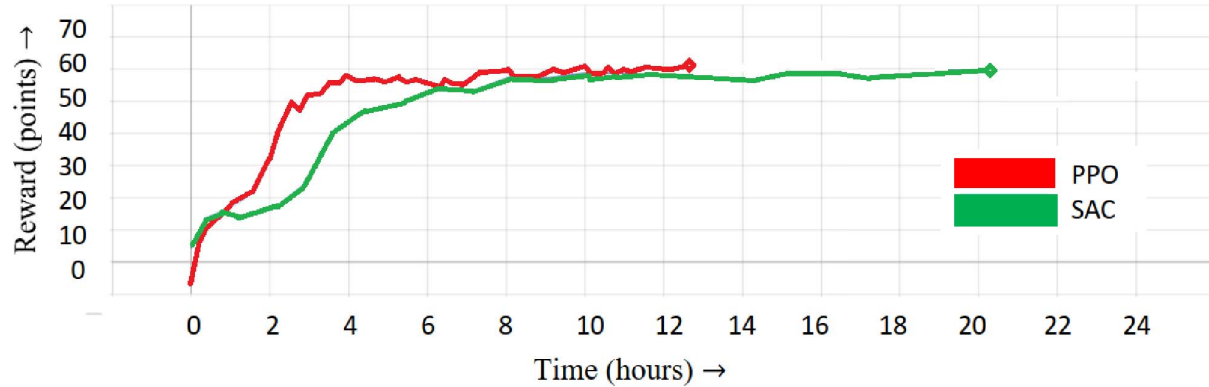


Figure 5: For each algorithm, the average episode awards against time-in-hour

F. Future Works

This trained agent performs in the case of lane change and lane merge towards avoiding the multiple vehicle collisions, but it does not mean agent is able to perform in others complicated scenarios. However, this might be encouraged by parallel implementation of the agent in several environments. According to the reality, this simulation is over-simplifying. Computer vision tasks are crucial difficulty for understanding the autonomous driving environment, in this simulation that is mostly ignored. Another crucial issue is weather conditions that can complicate the recognition of their surroundings by the artificial intelligence, although those conditions are always excellent in current simulations which make it much easier. To mimic a more realistic road conditions, future simulation research should include these conditions.

V. CONCLUSIONS

Towards avoid the multiple vehicle collision in autonomous driving, it is very important to deploy the optimum driving policy and both the PPO and SAC can be used to solution this problem with a reasonable success rate. While SAC demonstrated greater sampling efficiency than PPO, the SAC handled the same quantity of data as PPO, which was less optimum and more stable. The algorithms were not perfect, because they even had a slight performance error. This performance deficiency can be linked to a lack of processing power and a simpler learning process. Potential studies in future should use more realistic simulations in order to better represent and generalize results. The agent can also learn same things from different driving scenarios including lane changing circumstances while also using parallel simulations.

VI. ACKNOWLEDGMENT

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